Interface Definition Specification

Operator Tablet <-> Swarm — 35 Standard Messages

Interface Definition Specification (IDS)

문서번호 SAN-IDS-CMD-001

v1.1

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Classification	Interface Definition Specification
Preceding Docs	SAN-SDD-SWARM-001 v1.1
Message Format	Google Protobuf v3 + ROS 2 .msg
Transport	Wi-Fi 6 Mesh (OpenWrt) + Military LTE + LoRa
Message Count	35 (v1.0: 31 + 4 new + 2 modified)

Revision History

Version	Changes
v1.0	Initial release. 31 standard messages (8 commands + 9 telemetry + 9 inter-robot + 5 infra).
v1.1	All 31 v1.0 messages preserved with the following changes: (a) Added VideoStreamRequest in §3 (Android pull-mode video request); (b) Modified VideoStreamHandle in §4 to specify GStreamer SRT/UDP; (c) Added SurveillanceSectorAssignment, PanTiltCommand, SLAMAggregatedMap in §5 (3 new); (d) Replaced SLAMDelta with SLAMLocalDelta in §5 (collection period 1Hz -> 5s (v1.3 shortened)). Total 35 messages.
v1.4	Deputy UGV message fields + new messages: (1) LeaderRoleAnnouncement message new (Section 5.15); (2) HubRoleAnnouncement message new (Section 5.16); (3) RobotStatus gains is_deputy_ugv, battery_percent fields; (4) deputy_chain default updated — [3, 2, ...] (Deputy 1st).

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1. Overview

1.1 Document Scope

This document defines every communication message of the SAN swarm system in a single specification. Scope includes:

1. Operator Android app -> Leader (command issuance)
2. Leader -> Operator Android app (state/telemetry)
3. Leader -> Follower (formation, predicted poses, surveillance sectors)
4. Follower -> Leader/Hub (state, SLAM aggregated map)
5. Hub UGV <-> wide-area C2 (military LTE backhaul)

1.2 Message Categories

Category	Count	Direction
§3 Operator Commands	9 (+1 new)	Tablet -> Leader
§4 Swarm Telemetry	9 (1 mod)	Swarm -> Tablet
§5 Inter-Robot	12 (+3 new, 1 mod)	Leader <-> Follower <-> Hub
§6 Infrastructure/System	5	Bidirectional
Total	35	—

2. Message Architecture

2.1 Topic Naming Convention

```
     /{robot_namespace}/{category}/{message_type}      Examples:
/leader/cmd/formation                   # operator -> leader
/leader/cmd/surveillance_sectors         # operator -> leader (sector config)
/swarm/predict/follower_target           # leader broadcast
/swarm/surveillance/sector_assign        # leader -> followers (10s period)
/hub/slam/aggregated_map                 # Hub broadcast (5s (v1.3 shortened))
/{robot_id}/cmd/pantilt                   # leader or operator -> robot
/tablet/cmd/video_request                 # tablet -> hub
```

2.2 Namespace Policy

Namespace	Purpose	Example Topic
/leader	Leader (Go2) exclusive	/leader/status, /leader/cmd
/hub	Hub UGV exclusive	/hub/slam, /hub/video_forward
/follower{N}	Each follower (N=1..7)	/follower1/status
/swarm	All-broadcast	/swarm/predict, /swarm/surveillance
/tablet	Operator app	/tablet/heartbeat

2.3 QoS 4-Tier

Priority	Reliability	Durability	History	Deadline	Example
P0 Critical	RELIABLE	TRANSIENT_LOCAL	KEEP_LAST(10)	100 ms	FormationCommand, EmergencyStop
P1 Tactical	BEST_EFFORT/REL	VOLATILE	KEEP_LAST(3)	200 ms	RobotStatus, FollowerTarget
P2 Operational	RELIABLE	TRANSIENT_LOCAL	KEEP_LAST(5)	1 s	SLAMLocalDelta, SLAMAggregatedMap
P3 Background	BEST_EFFORT	VOLATILE	KEEP_LAST(1)	5 s	LogEvent

3. Operator Command Messages (Tablet -> Leader/Hub)

3.1 FormationCommand

Attribute	Value
Direction	Tablet -> Leader
Topic	/leader/cmd/formation
QoS	P0, RELIABLE, KEEP_LAST(10)
Rate	Event

```
     message FormationCommand {         enum FormationType {             COLUMN=0;
LINE=1; V_SHAPE=2; DIAMOND=3;             ECHELON_LEFT=4; ECHELON_RIGHT=5; BOX=6;
VEE_INVERTED=7; FREE_SPREAD=8;         }         FormationType  type             = 1;
float          d                = 2;   // 1.5 <= d <= 20         float
theta_deg        = 3;          float          alpha_deg        = 4;          float
transition_time  = 5;          bool           dev_mode         = 6;          uint64
timestamp_ms     = 7;          string         requested_by     = 8;     }
```

3.2 MissionStateCommand

```
     message MissionStateCommand {         enum State {             IDLE=0; ASSEMBLY=1;
OFFENSIVE=2; DEFENSIVE=3;             PATROL=4; RTH=5; HOLD=6; EMERGENCY=7;         }
State    target_state     = 1;         Pose2D   target_position  = 2;         uint64
timestamp_ms     = 3;          string   requested_by     = 4;     }     // QoS: P0,
RELIABLE
```

3.3 FireAuthorization

```
     message FireAuthorization {         string    target_id           = 1;
uint32    assigned_robot       = 2;         enum FireMode {             SINGLE_SHOT=0;
THREE_BURST=1; FULL_AUTO=2; JAMMING=3;         }         FireMode  mode
= 3;         Pose2D    target_pose         = 4;         float     lead_distance_m     =
5;         uint64    expires_at_ms       = 6;         bytes     operator_signature  =
7;         uint64    timestamp_ms        = 8;     }     // QoS: P0, RELIABLE, KEEP_ALL
```

3.4 JammingCommand

```
     message JammingCommand {         enum Band {             BAND_2_4_GHZ=0;
BAND_5_8_GHZ=1;             BAND_GNSS_L1=2; BAND_GNSS_L2=3; BAND_ALL=4;         }
repeated Band  bands            = 1;         Pose2D         target_direction  = 2;
float          duration_s       = 3;         float          power_w           = 4;
uint64         timestamp_ms     = 5;     }     // QoS: P0, RELIABLE
```

3.5 WaypointCommand

```
     message WaypointCommand {         repeated Pose2D  waypoints         = 1;
float            cruise_speed_mps  = 2;   // 0=auto, max 1.5         bool
allow_replan      = 3;         uint32           priority          = 4;         uint64
timestamp_ms      = 5;     }     // QoS: P1, RELIABLE
```

3.6 ManualOverrideCommand

```
    message ManualOverrideCommand {        string  robot_id        = 1;        enum
Mode {            DISENGAGE_AUTO=0; DIRECT_DRIVE=1; HOLD_POSITION=2; FOLLOW_ME=3;
}        Mode    mode             = 2;        Velocity2D  manual_vel    = 3;
uint64  expires_at_ms    = 4;        uint64  timestamp_ms     = 5;     }     // QoS:
P0, RELIABLE
```

3.7 EmergencyStop

```
    message EmergencyStop {        enum Severity {            SOFT_STOP=0;
HARD_STOP=1; DISARM=2; SHUTDOWN=3;        }        Severity  severity         = 1;
string    reason           = 2;        bytes     auth_signature   = 3;        uint64
timestamp_ms     = 4;     }     // QoS: P0, RELIABLE, KEEP_ALL, LoRa fallback ON
```

3.8 OperatorHeartbeat

```
    message OperatorHeartbeat {        string    operator_id      = 1;        bytes
auth_token       = 2;        Pose2D    operator_pose    = 3;        uint32
app_version      = 4;        uint64    timestamp_ms     = 5;     }     // QoS: P1,
BEST_EFFORT, 1 Hz
```

3.9 VideoStreamRequest (v1.1 new)

Pull-mode request issued by the Android app. Prevents unnecessary bandwidth and power consumption.

Attribute	Value
Direction	Tablet -> Hub UGV (SBC #2)
Topic	/tablet/cmd/video_request
QoS	P1, RELIABLE
Rate	Event (operator taps View Video)

```
    message VideoStreamRequest {        uint32  sequence           = 1;
uint32  target_robot_id     = 2;        string  protocol           = 3;   // "srt" |
"udp"        uint32  desired_port        = 4;        string  codec              = 5;
// "h265" | "h264"        string  quality             = 6;   // "hd" | "fhd" | "low" |
"thumbnail"        bool    encryption          = 7;        string  passphrase_hint
= 8;        string  action              = 9;   // "start" | "stop" | "change_quality"
uint64  timestamp_ms        = 10;     }
```

4. Swarm Telemetry Messages (Swarm -> Tablet)

4.1 RobotStatus

```
    message RobotStatus {         string  robot_id           = 1;         enum Role
{ LEADER=0; HUB=1; FOLLOWER=2; }         Role    role               = 2;         Pose2D
pose               = 3;         Velocity2D  velocity       = 4;         float
heading_rad        = 5;         enum Tier {             T0_PREDICTIVE=0;
T1_5_AUTO_REROUTE=1; T1_NORMAL=2;             T2_CATCH_UP=3; T3_HARD_CATCH_UP=4;
T4_BREADCRUMB=5;         }         Tier    current_tier       = 6;         float
battery_percent    = 7;         float   battery_voltage    = 8;         float
cpu_load_percent   = 9;         float   mem_used_percent   = 10;         bool
rtk_fixed          = 11;         bool    comm_healthy       = 12;         bool
lte_active         = 13;   // v1.3: this robot is the active LTE gateway         bool
is_lte_backup_designated = 14;  // v1.3: designated as LTE backup follower
uint32  errors_active      = 13;         uint64  timestamp_ms       = 14;     }     //
QoS: P1, BEST_EFFORT, 1 Hz
```

4.2 FormationStatus

```
    message FormationStatus {         FormationType  active_type        = 1;
float          d                  = 2;         float          theta_deg          = 3;
MissionState   mission_state      = 4;         uint32         total_robots       = 5;
uint32         healthy_robots     = 6;         repeated string  unhealthy_ids    = 7;
float          formation_error_m  = 8;         bool           auto_column_active = 9;
uint64         timestamp_ms       = 10;     }     // QoS: P1, BEST_EFFORT, 2 Hz
```

4.3 DetectedObject

```
    message DetectedObject {         string  object_id          = 1;         string
detector_robot_id  = 2;         enum ObjectClass {             PERSON=0; VEHICLE=1;
DRONE_FPV=2; DRONE_MULTI=3;             DRONE_FIXED=4; UNKNOWN=5;         }
ObjectClass  obj_class    = 3;         float        confidence   = 4;         Pose2D
position     = 5;         Velocity2D   velocity_estimate = 6;         float
range_m      = 7;         float        bearing_deg  = 8;         bool         is_threat
= 9;         uint32       threat_score = 10;         bytes        thumbnail_jpeg = 11;
uint64       detected_at_ms = 12;     }     // QoS: P0, RELIABLE, event
```

4.4 ThreatAlert

```
    message ThreatAlert {         string  alert_id          = 1;         enum Severity
{             ADVISORY=0; CAUTION=1; WARNING=2; CRITICAL=3;         }         Severity
severity         = 2;         enum ThreatType {             DRONE_APPROACHING=0;
ENEMY_PERSONNEL=1; ARTILLERY_DETECTED=2;             AMBUSH_INDICATOR=3;
COMM_JAMMING=4; GPS_JAMMING=5;         }         ThreatType  threat_type    = 3;
string  related_object_id  = 4;         float   time_to_impact_s   = 5;         Pose2D
threat_position    = 6;         string  message_text       = 7;         uint64
timestamp_ms       = 8;     }     // QoS: P0, RELIABLE, KEEP_ALL
```

4.5 TierStatusChange

```
    message TierStatusChange {         string    robot_id          = 1;         Tier
from_tier         = 2;         Tier      to_tier           = 3;         string
```

```
reason             = 4;         float     delta_m           = 5;         uint64
timestamp_ms       = 6;     }     // QoS: P0, RELIABLE, event
```

4.6 FireResult

```
     message FireResult {         string  fire_id              = 1;         string
shooter_robot_id     = 2;         string  target_object_id     = 3;         uint32
rounds_fired         = 4;         bool    target_neutralized   = 5;         float
hit_probability      = 6;         bytes   post_image_jpeg      = 7;         uint64
fired_at_ms          = 8;     }     // QoS: P0, RELIABLE
```

4.7 BatteryWarning

```
     message BatteryWarning {         string  robot_id           = 1;         enum
Level { LOW=0; CRITICAL=1; SHUTDOWN=2; }         Level   level              = 2;
float   percent            = 3;         uint32  minutes_remaining  = 4;         bool
should_rth         = 5;         uint64  timestamp_ms       = 6;     }     // QoS: P0,
RELIABLE
```

4.8 VideoStreamHandle (v1.1 modified — GStreamer specified)

Hub UGV returns video stream handle to APP. v1.0's simple RTSP URL is now extended to SRT/UDP URI + encryption key + actual bitrate.

```
     message VideoStreamHandle {         uint32  sequence             = 1;
uint32  target_robot_id      = 2;         string  srt_uri              = 3;   // e.g.
"srt://hub_ip:8888?mode=listener"         string  codec                = 4;   // "h265"
| "h264"         string  quality              = 5;   // "hd" | "fhd" | "low" |
"thumbnail"         uint32  actual_bitrate_kbps = 6;         string  passphrase
= 7;         string  status               = 8;   // "streaming" | "stopped" | "error"
string  error_msg            = 9;         uint64  stream_start_ms      = 10;
uint64  timestamp_ms         = 11;     }     // QoS: P1, RELIABLE
```

4.9 SLAMUpdate

```
     message SLAMUpdate {         Pose2D  origin               = 1;         float
resolution_m        = 2;         uint32  width_cells         = 3;         uint32
height_cells         = 4;         bytes   occupancy_delta      = 5;         uint32
contributing_robots = 6;         uint64  timestamp_ms        = 7;     }     // QoS: P2,
RELIABLE, 0.5 Hz (for tablet screen refresh)
```

5. Inter-Robot Messages

5.1 FollowerTargetMessage (1s prediction)

Attribute	Value
Direction	Leader -> Follower (broadcast)
Topic	/swarm/predict/follower_target
QoS	P0, RELIABLE, KEEP_LAST(3)
Rate	10 Hz (100 ms)
Size	~200 byte

```
     message FollowerTargetMessage {         uint32  follower_id            = 1;
Pose2D  target_pose            = 2;         uint64  valid_at_ms            = 3;
repeated Pose2D  leader_path_segment = 4;         uint32  sequence_num           = 5;
float   confidence             = 6;         Velocity2D  target_velocity    = 7;
uint64  generated_at_ms        = 8;     }      message Pose2D {         double  x
= 1;         double  y         = 2;         float   heading  = 3;     }     message
Velocity2D {         float   vx       = 1;         float   vy       = 2;         float
omega    = 3;     }
```

5.2 BreadcrumbBroadcast

```
     message BreadcrumbBroadcast {         uint64  timestamp_us       = 1;
Pose2D  position           = 2;         float   speed_mps          = 3;         enum
TerrainClass {             OPEN=0; FOREST=1; URBAN=2; NARROW=3; CLIFF=4;         }
TerrainClass  terrain        = 4;         uint8   formation_id       = 5;
uint32  sequence_num       = 6;     }     // QoS: P1, RELIABLE, KEEP_LAST(50), 0.5-2 Hz
(1m moved or 0.5s)
```

5.3 LeaderElection (Modified Raft)

```
     message LeaderElection {         enum MessageType {             REQUEST_VOTE=0;
VOTE_GRANTED=1; VOTE_DENIED=2;             LEADER_ANNOUNCE=3; HEARTBEAT=4;         }
MessageType  type                = 1;         uint32       term                = 2;
string       candidate_id        = 3;         float        candidate_priority  =
4;   // is_hub_ugv top priority         string       voter_id             = 5;
uint64       timestamp_ms        = 6;     }     // QoS: P0, RELIABLE, KEEP_ALL, event
(Hub UGV is priority 1)
```

5.4 EmergencyAlert

```
     message EmergencyAlert {         string  source_robot_id  = 1;         enum
AlertType {             OBSTACLE_AHEAD=0; COLLISION_WARNING=1; CLIFF_DETECTED=2;
FORMATION_BREAK=3; SENSOR_FAIL=4;         }         AlertType  type             = 2;
Pose2D     alert_position   = 3;         float      severity         = 4;
string     description      = 5;         uint64     timestamp_ms     = 6;     }     //
QoS: P0, RELIABLE, event
```

5.5 SLAMLocalDelta (v1.1 modified — 1Hz -> 5s (v1.3 shortened) aggregated)

Replaces the v1.0 SLAMDelta (1 Hz). Sends an accumulated aggregated map every 5s (v1.3 shortened), reducing comm load by 1/100.

```
     message SLAMLocalDelta {         uint32  sequence                  = 1;
string  robot_id                  = 2;         bytes   occupancy_grid_delta_png  = 3;
// PNG compressed         Pose2D  origin                    = 4;         float
resolution_m              = 5;         uint64  coverage_start_ms         = 6;   //
aggregation start time         uint64  coverage_end_ms           = 7;   // aggregation
end time         uint64  timestamp_ms              = 8;     }     // QoS: P2, RELIABLE
// Direction: all robots (Leader + 7 Follower)     // Receiver: Hub UGV SBC #1     //
Rate: 5s (v1.3 shortened) (default 30s, dynamic adjustment)
```

5.6 SLAMAggregatedMap (v1.1 new)

Broadcast of the merged global map by Hub UGV SBC #1. Updates the Nav2 cost map static layer on every robot.

```
     message SLAMAggregatedMap {         uint32  sequence            = 1;         bytes
occupancy_grid_png  = 2;   // PNG compressed global map         Pose2D  origin
= 3;         float   resolution_m        = 4;         uint32  width_cells         = 5;
uint32  height_cells        = 6;         uint32  contributing_robots = 7;
uint64  timestamp_ms        = 8;     }     // QoS: P2, RELIABLE, TRANSIENT_LOCAL     //
Sender: Hub UGV (SBC #1)     // Receiver: all robots + operator tablet     // Rate: 5s
(v1.3 shortened) (dynamic)
```

5.7 GlobalMapBroadcast (v1.0 retained, operator screen)

```
     message GlobalMapBroadcast {         Pose2D  map_origin          = 1;
float   resolution_m        = 2;         uint32  width               = 3;
uint32  height              = 4;         bytes   full_map            = 5;         bytes
delta_map           = 6;         repeated string contributor_ids = 7;         uint64
timestamp_ms        = 8;     }     // QoS: P2, RELIABLE, TRANSIENT_LOCAL, 0.5 Hz
```

5.8 HeartBeat

```
     message HeartBeat {         string  robot_id          = 1;         uint32
sequence_num      = 2;         uint64  uptime_ms         = 3;         float
cpu_load_percent  = 4;         bool    is_leader         = 5;         uint64
timestamp_ms      = 6;     }     // QoS: P1, BEST_EFFORT, 1 Hz
```

5.9 SlotAssignment

```
     message SlotAssignment {         string  follower_id        = 1;         Pose2D
offset_in_body     = 2;         uint32  transition_id      = 3;         float
transition_time_s  = 4;         uint32  priority           = 5;         uint64
valid_from_ms      = 6;         uint64  timestamp_ms       = 7;     }     // QoS: P0,
RELIABLE, event (on formation change, Hungarian result)
```

5.10 SurveillanceSectorAssignment (v1.1 new)

Leader assigns surveillance sectors to each follower. Published every 10s (default) or immediately on events.

Attribute	Value
Direction	Leader -> all Followers + Hub UGV (broadcast)
Topic	/swarm/surveillance/sector_assign
QoS	P1, RELIABLE
Rate	10s (periodic) + event (formation change, threat)

```
    message SurveillanceSectorAssignment {        uint32  sequence              = 1;
uint32  robot_id             = 2;         // target         float   sector_start_deg
= 3;          // -180 to 180 (heading-based)        float   sector_end_deg        = 4;
uint32  valid_period_sec     = 5;         // default 10         string  priority
= 6;          // "primary" | "threat_focus" | "overlap"         string  mode_hint
= 7;          // "sweep" | "track" | "fixed"        uint64  timestamp_ms          = 8;
}
```

5.11 PanTiltCommand (v1.1 new)

Leader or operator controls a robot's pan-tilt. Followers may also self-publish to start sweep upon receiving sector assignment.

Attribute	Value
Direction	Leader or Tablet -> Robot
Topic	/{robot_id}/cmd/pantilt
QoS	P1, RELIABLE
Rate	Event or on sweep entry

```
    message PanTiltCommand {        uint32  sequence            = 1;         uint32
robot_id            = 2;         float   target_pan_deg      = 3;          // heading-
based         float   target_tilt_deg     = 4;         float   speed_dps            = 5;
// degrees/sec         string  mode                = 6;          // "sweep" | "track"
| "fixed" | "engage"         float   sweep_range_deg     = 7;           // sweep width
uint64  timestamp_ms        = 8;     }
```

5.12 SwarmHealthSummary

```
    message SwarmHealthSummary {        uint32  total_robots          = 1;
uint32  online_robots         = 2;         float   avg_battery_percent   = 3;
float   min_battery_percent   = 4;         float   avg_formation_err_m   = 5;
float   comm_quality_percent  = 6;         uint32  robots_in_t4          = 7;
uint32  robots_in_emergency   = 8;         uint64  timestamp_ms          = 9;     }
// QoS: P1, BEST_EFFORT, 1 Hz, Hub -> Tablet
```

5.13 CostMapUpdate (v1.3 new)

Each UGV publishes its computed Local Cost Map. Hub UGV SBC #1 collects them for cross-validation.

Attribute	Value
Direction	Each UGV -> broadcast
Topic	/{robot_id}/local/cost_map
QoS	P2, BEST_EFFORT, KEEP_LAST(1)
Rate	1 Hz (KPP: cost_map_latency_p99 <= 5s)

5.14 LTERoleAnnouncement (v1.3 new)

LTE gateway promotion/demotion broadcast. When Hub UGV (S2) LTE fails, Follower S3 (1st priority) auto-promotes.

Attribute	Value
Direction	LTE backup promoting/demoting robot -> broadcast
Topic	/swarm/lte/role_announce
QoS	P0, RELIABLE, KEEP_ALL
Rate	Event (on promotion/demotion/re-affirm)

6. Infrastructure/System Messages

6.1 TopicCatalog

```
     message TopicCatalog {         string  robot_id           = 1;         repeated
TopicEndpoint  publishers   = 2;         repeated TopicEndpoint  subscribers  = 3;
uint64  generated_at_ms    = 4;     }     message TopicEndpoint {         string
topic_name      = 1;         string  message_type    = 2;         string  qos_profile
= 3;     }     // QoS: P1, RELIABLE, TRANSIENT_LOCAL, event (startup/change)
```

6.2 VersionNegotiation

```
     message VersionNegotiation {         string         peer_id             = 1;
uint32         schema_major        = 2;         uint32         schema_minor        = 3;
uint32         sw_version          = 4;         repeated uint32 supported_majors   = 5;
bool           is_compatible       = 6;         string         error_message       = 7;
uint64         timestamp_ms        = 8;     }     // QoS: P0, RELIABLE
```

6.3 SecurityHandshake

```
    message SecurityHandshake {         string  peer_id              = 1;
bytes   certificate_chain    = 2;   // X.509 v3         bytes   nonce                =
3;         bytes   signature            = 4;   // RSA 4096         string  cipher_suite
= 5;   // "AES-256-GCM"         uint64  timestamp_ms         = 6;     }     // QoS: P0,
RELIABLE
```

6.4 LogEvent

```
    message LogEvent {         string  source_id        = 1;         enum Severity {
DEBUG=0; INFO=1; WARN=2; ERROR=3; FATAL=4;         }         Severity  severity
= 2;         string  component        = 3;         string  message          = 4;
string  correlation_id   = 5;         uint64  timestamp_ms     = 6;     }     // QoS:
P3, BEST_EFFORT
```

6.5 TimeSync

```
    message TimeSync {         uint64  master_time_us     = 1;         int64
offset_us          = 2;         float   drift_ppm          = 3;         uint32
sync_round         = 4;     }     // QoS: P0, BEST_EFFORT, 1 Hz, Hub master -> all
robots     // PTP IEEE-1588 grandmaster on Hub UGV SBC #1
```

7. Security (DDS-Security)

Function	Standard	Adopted Value
Authentication	OMG DDS-Security AuthN	PKI X.509 v3, RSA 4096
Encryption	OMG DDS-Security Crypto	AES-256-GCM
Access Control	OMG DDS-Security AccessCtrl	Permissions Document
Logging	OMG DDS-Security Logging	Centralized on Hub UGV
Key Management	Internal policy	Hub UGV TPM, per-mission rotation

7.1 Video Stream Security

- SRT built-in AES-128 encryption (per-mission passphrase)
- Hub-Operator segment: IPsec or OpenVPN tunnel (additional)
- Robot-Hub LAN: WPA3-SAE + DDS Security

7.2 Emergency Key Revocation

On suspected robot loss/capture, the operator app's "Compromise" button immediately revokes that robot's certificate. The CRL is broadcast to the entire swarm immediately.

8. Appendix — Message Catalog (35 total)

8.1 Command Messages (9)

#	Name	Topic	QoS	Rate	v1.1
3.1	FormationCommand	/leader/cmd/formation	P0	evt	
3.2	MissionStateCommand	/leader/cmd/mission_state	P0	evt	
3.3	FireAuthorization	/leader/cmd/fire_auth	P0	evt	
3.4	JammingCommand	/leader/cmd/jamming	P0	evt	
3.5	WaypointCommand	/leader/cmd/waypoint	P1	evt	
3.6	ManualOverrideCommand	/{robot}/cmd/manual_override	P0	evt	
3.7	EmergencyStop	/swarm/emergency_stop	P0	evt	
3.8	OperatorHeartbeat	/leader/heartbeat/operator	P1	1 Hz	
3.9	VideoStreamRequest	/tablet/cmd/video_request	P1	evt	* new

8.2 Telemetry Messages (9)

#	Name	Topic	QoS	Rate	v1.1
4.1	RobotStatus	/{robot}/status/robot	P1	1 Hz	
4	FormationStatus	/leader/status/formation	P1	2 Hz	

#	Name	Topic	QoS	Rate	v1.1
.2					
4.3	DetectedObject	/{robot}/detected/object	P0	evt	
4.4	ThreatAlert	/leader/alert/threat	P0	evt	
4.5	TierStatusChange	/{robot}/status/tier_change	P0	evt	
4.6	FireResult	/{robot}/status/fire_result	P0	evt	
4.7	BatteryWarning	/{robot}/warn/battery	P0	evt	
4.8	VideoStreamHandle	/leader/status/video_handle	P1	evt	* mod
4.9	SLAMUpdate	/hub/slam/global_map_delta	P2	0.5 Hz	

8.3 Inter-Robot Messages (12)

#	Name	Topic	QoS	Rate	v1.1
5.1	FollowerTargetMessage	/swarm/predict/follower_target	P0	10 Hz	
5.2	BreadcrumbBroadcast	/swarm/breadcrumb	P1	0.5-2 Hz	
5.3	LeaderElection	/swarm/leader_election	P0	evt	
5.	EmergencyAlert	/swarm/alert/emergency	P0	evt	

#	Name	Topic	QoS	Rate	v1.1
4					
5.5	SLAMLocalDelta	/{robot}/slam/local_delta	P2	5s (v1.3 shortened)	* mod
5.6	SLAMAggregatedMap	/hub/slam/aggregated_map	P2	5s (v1.3 shortened)	* new
5.7	GlobalMapBroadcast	/swarm/slam/global_map	P2	0.5 Hz	
5.8	HeartBeat	/{robot}/heartbeat	P1	1 Hz	
5.9	SlotAssignment	/{follower}/cmd/slot_assignment	P0	evt	
5.10	SurveillanceSectorAssignment	/swarm/surveillance/sector_assign	P1	10 s	* new
5.11	PanTiltCommand	/{robot}/cmd/pantilt	P1	evt	* new
5.12	SwarmHealthSummary	/hub/health/swarm_summary	P1	1 Hz	

8.4 Infrastructure/System Messages (5)

#	Name	Topic	QoS	Rate	v1.1
6	TopicCatalog	/{robot}/system/	P1	evt	

#	Name	Topic	QoS	Rate	v1.1
.1		topic_catalog			
6.2	VersionNegotiation	/swarm/system/version_neg	P0	evt	
6.3	SecurityHandshake	/swarm/system/security_handshake	P0	evt	
6.4	LogEvent	/{robot}/system/log	P3	evt	
6.5	TimeSync	/swarm/system/time_sync	P0	1 Hz	

8.5 v1.0 -> v1.1 Change Summary

Change	Messages	Rationale
New (+4)	VideoStreamRequest, SurveillanceSectorAssignment, PanTiltCommand, SLAMAggregatedMap	v1.1 four design changes
Modified (2)	VideoStreamHandle (GStreamer SRT specified), SLAMLocalDelta (replaces SLAMDelta, 5s (v1.3 shortened))	Protocol specification + period adjustment
Retained (29)	All other v1.0 messages	—
Total	35 messages	—